Convex Stein domains are absolute localizations

A positive solution of the open unit-ball case

Autonomous solve-first investigation, lane 12

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Abstract

Aristov and Pirkovskii ask whether restriction to a Stein open subspace induces a strong homological epimorphism, and state that this is unknown even for the unit ball in $\mathbb{C}^2$. We answer that case positively and prove more: for every convex domain $U \subset \mathbb{C}^n$, the restriction $\mathcal{O}(\mathbb{C}^n) \to \mathcal{O}(U)$ is a strong homological epimorphism (Taylor's absolute localization). The proof base-changes the finite Koszul resolution of the diagonal. Convexity supplies an explicit continuous contracting homotopy on the resulting holomorphic Koszul complex: in difference coordinates $v = z - w$, it is $d_v(E + k)^{-1}$ on exterior degree k. No boundary regularity or boundedness is needed. The general nonconvex Stein question is not claimed.

1 Statement and homological reduction

All tensor products below are completed projective tensor products. A Fréchet algebra homomorphism $A \to B$ is a *strong homological epimorphism* if, after applying $B\widehat{\otimes}_A(-)\widehat{\otimes}_A B$ to a projective A-bimodule resolution of A, the resulting augmented complex is admissible, i.e. split exact as a complex of Fréchet spaces [1].

Theorem 1. *Let $U \subset \mathbb{C}^n$ be a nonempty convex domain. Then restriction*

$$\rho_U : \mathcal{O}(\mathbb{C}^n) \longrightarrow \mathcal{O}(U)$$

is a strong homological epimorphism.

Corollary 2. *For the open Euclidean unit ball $B_n \subset \mathbb{C}^n$, and in particular for the case $n = 2$ singled out in [1, Question 6.1], $\mathcal{O}(\mathbb{C}^n) \to \mathcal{O}(B_n)$ is a strong homological epimorphism.*

Put $A = \mathcal{O}(\mathbb{C}^n)$ and let $E = \mathbb{C}^n$ with basis $e_1, \dots, e_n$. Under the nuclear kernel isomorphism

$$A^e = A\widehat{\otimes}A \cong \mathcal{O}(\mathbb{C}^n \times \mathbb{C}^n),$$

the standard finite free A-bimodule resolution of A is the augmented Koszul complex

$$0 \longrightarrow A^e \otimes \Lambda^n E \longrightarrow \cdots \longrightarrow A^e \otimes E \xrightarrow{\delta} A^e \xrightarrow{\epsilon} A \longrightarrow 0, \tag{1}$$

where $\epsilon f(w) = f(w, w)$ and

$$\delta(f\, e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{r=1}^{k} (-1)^{r-1}(z_{i_r} - w_{i_r}) f\, e_{i_1} \wedge \cdots \widehat{e}_{i_r} \cdots \wedge e_{i_k}.$$

For completeness, the contracting homotopy below with $U = \mathbb{C}^n$ proves that (1) is admissible; its terms are finite free A^e-modules.

Let $B = \mathcal{O}(U)$. Explicit cancellation of the balanced A-actions gives

$$B\widehat{\otimes}_A(A^e \otimes \Lambda^k E)\widehat{\otimes}_A B \cong (B\widehat{\otimes}B) \otimes \Lambda^k E \cong \mathcal{O}(U \times U) \otimes \Lambda^k E.$$

Thus the base change of (1) is the same diagonal Koszul complex on $U \times U$. It remains only to prove that this complex has a continuous contraction.

2 The radial Koszul contraction

Use the linear coordinates $(v, w) = (z - w, w)$ and write

$$D_U = \{(v, w) \in \mathbb{C}^n \times \mathbb{C}^n : w \in U,\ w + v \in U\}.$$

Convexity says that

$$(v, w) \in D_U, \quad 0 \le t \le 1 \quad \Longrightarrow \quad (tv, w) \in D_U. \tag{2}$$

Identify a Koszul k-chain with a holomorphic exterior k-form in the formal basis e_j. Let

$$d_v = \sum_{j=1}^{n} e_j \wedge \frac{\partial}{\partial v_j}, \qquad E = \sum_{j=1}^{n} v_j \frac{\partial}{\partial v_j},$$

and let $\delta = \iota_v$ be contraction by v. On exterior degree k,

$$\delta d_v + d_v \delta = E + k. \tag{3}$$

For $k \ge 1$ define a continuous linear operator

$$(R_k \omega)(v, w) = \int_0^1 t^{k-1} \omega(tv, w)\, dt, \tag{4}$$

and for functions put

$$(R_0 f)(v, w) = \int_0^1 \frac{f(tv, w) - f(0, w)}{t}\, dt. \tag{5}$$

The integrand in (5) has a removable singularity at $t = 0$. Differentiating $t^k \omega(tv, w)$ gives

$$(E + k) R_k \omega = \omega \quad (k \ge 1), \qquad E R_0 f = f - Pf, \tag{6}$$

where $(Pf)(v, w) = f(0, w)$. Also

$$R_{k-1} \delta = \delta R_k \qquad (k \ge 1); \tag{7}$$

for $k = 1$, this uses $(\delta\omega)(0, w) = 0$.

Set $h_k = d_v R_k$. Combining (3)–(7) yields

$$\delta h_k + h_{k-1} \delta = I \quad (k \ge 1), \qquad \delta h_0 = I - P. \tag{8}$$

Finally, $s : \mathcal{O}(U) \to \mathcal{O}(D_U)$, $(sg)(v, w) = g(w)$, satisfies $\epsilon s = I$ and $s\epsilon = P$. Hence (8) contracts the entire augmented Koszul complex.

We record the topological point. If $K \Subset D_U$, then by (2)

$$K^\star = \{(tv, w) : (v, w) \in K,\ 0 \le t \le 1\}$$

is a compact subset of D_U. For $k \ge 1$, compact-open seminorms of $R_k \omega$ on K are controlled by seminorms of ω on $K^\star$. For R_0, use

$$\frac{f(tv, w) - f(0, w)}{t} = \int_0^1 \sum_{j=1}^{n} v_j \frac{\partial f}{\partial v_j}(stv, w)\, ds;$$

boundedness of v on K and continuity of holomorphic differentiation give the corresponding seminorm estimate. Hence every R_k and $h_k = d_v R_k$ is continuous. Thus the augmented complex is split exact as a complex of Fréchet spaces. By the base-change reduction above, this is precisely the strong homological epimorphism property and proves Theorem 1.

3 Scope

The proof actually isolates a useful sufficient mechanism: a continuous contraction of the diagonal Koszul complex after restriction. Convexity provides one by the elementary radial formula, so the result covers unbounded convex domains and requires no boundary regularity.

For a general Stein domain, Hefer division and Cartan theory give algebraic exactness of related complexes, but strongness asks for continuous linear splittings in the Fréchet topology. Exact sequences of nuclear Fréchet spaces need not split, and no global replacement for (2) is provided here. Therefore the full Stein-open-subspace question and the source's questions on singular C^∞-differentiable spaces remain open.

Eight focused upgrade routes were tested. Bounded exact-question, unit-ball, absolute-localization, Koszul, Hefer, corrigendum, and citation searches on 13 August 2026 found no later resolution of the unit-ball case or the convex-domain theorem. Novelty is plausible, not certified.

References

[1] O. Yu. Aristov and A. Yu. Pirkovskii, *Open embeddings and pseudoflat epimorphisms*, J. Math. Anal. Appl. 485 (2020), 123817; arXiv:1908.02117.